

Entanglement Swapping Sandbox

Technical Brief

Artifact class: Scenario-screening and technical decision-support tool for a two-source / one-hub entanglement-swapping model.

Purpose: Expose how source mismatch, Bell-state measurement misalignment, and loss affect branch success, remote-state quality, and useful heralded entanglement.

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Technical Summary

This brief presents the **Entanglement Swapping Sandbox**, a compact technical artifact for **scenario screening**, **tradeoff exploration**, and **technical specification support** in a minimal repeater-style model. The sandbox is built around two entanglement sources and one hub performing a Bell-state measurement, with adjustable source states, controllable Bell-state-measurement misalignment, simple loss, and branch-resolved remote-state metrics.

Core insight. Bell-state measurement success and useful entanglement swapping are not the same thing. A valid branch can occur while still producing a degraded remote state. The sandbox makes that distinction explicit by separating branch probability, valid probability after loss, and conditional swapped-state quality.

Classical channel and relativity. In the sandbox model, a Bell-state measurement produces one of four possible outcomes. In the ideal symmetric limit these occur with equal probability; in realistic settings (source mismatch, misalignment, loss) the branch probabilities are generally unequal. The classical channel communicates which outcome occurred — not to create entanglement, but to reveal which conditional entangled state the remote parties share. Without this classical message, the remote parties must average over outcomes, and the effective local description is a mixed state with no usable, known entanglement. It is precisely this classical communication bound by c that keeps entanglement swapping consistent with special relativity and prevents faster-than-light signaling. In a repeater architecture, classical-channel latency is therefore not overhead — it is load-bearing [1].

Companion document: see the matching **Implementation Brief** for theory, validation, software structure, and model limitations.

Problem Addressed

Entanglement swapping is a foundational primitive for repeater-style quantum networking. In principle, it allows two remote endpoints to become entangled without direct interaction. In practice, technical performance depends on several coupled factors:

- the quality and balance of the source states,
- the quality of the Bell-state measurement (BSM) at the hub,
- loss on the source-to-hub links,
- and the interpretation of branch success versus useful swapped entanglement.

These distinctions are often compressed into a single statement that “entanglement swapping was demonstrated.” While this phrasing may be sufficient for experts, cross-domain technical discussions often benefit from making the enabling factors more explicit:

Which combinations of source quality, BSM quality, and link loss produce remote entanglement that is not merely detectable, but technically useful?

What the Sandbox Models

The simulator focuses on a compact repeater-style abstraction:

- two adjustable entanglement sources,
- one hub performing Bell-state measurement,
- optional Bell-basis misalignment at the hub,
- simple source-to-hub loss,
- branch-resolved output metrics.

The source states can be tuned away from the ideal maximally entangled case, which supports direct study of source mismatch. The Bell-state measurement basis can also be rotated away from the ideal Bell basis to model imperfect alignment. Loss is included as a simple link-level factor on valid heralded event rate.

This makes the sandbox useful for rapid architecture-facing interpretation without requiring a full optical-circuit or end-to-end network stack.

Scenario Screening Value

The primary purpose of the sandbox is **scenario screening**. In this brief, scenario screening means using a compact model to answer questions such as:

- How sensitive is swapped entanglement to source imbalance?
- How much does BSM misalignment degrade remote-state quality?
- Which parameters most strongly affect useful heralded entanglement yield?
- Which operating regimes are promising enough to justify deeper modeling?

This is particularly useful in early-stage technical discussions, where a compact, inspectable artifact can help narrow the design space before more detailed simulation or experimental planning.

Scenario-screening function. The sandbox is not an ROI tool. It is a technical decision-support artifact for screening parameter regimes, exposing dominant sensitivities, and informing downstream requirements and specifications.

Representative Metrics

The simulator reports branch-resolved and summary metrics:

- branch probability,
- valid probability after loss,
- concurrence,
- negativity,
- best Bell-state fidelity,
- entanglement yield,
- fidelity yield.

These metrics support different interpretations:

- **Probability metrics** indicate how often a branch is expected.
- **Entanglement metrics** indicate how nonclassical the remote state remains.
- **Yield metrics** combine occurrence and quality, making them useful for screening and comparison.

Reporting multiple metrics at once helps avoid over-interpreting any single number in isolation. As a rough orientation: concurrence and negativity near unity indicate near-maximally entangled remote states; values below ~ 0.5 often indicate significant degradation for many use cases (thresholds are application-dependent). Fidelity yield combines occurrence rate and state quality into a single screening figure — a useful first filter before committing to deeper modeling or experimental planning.

Use Cases

The sandbox is aimed at technical audiences interested in entanglement swapping as an enabling element for quantum networking. Representative use cases include:

- **Technical stakeholder communication:** explain entanglement swapping without requiring a full simulation framework.
- **Architecture exploration:** compare sensitivity to source mismatch, BSM quality, and loss.
- **Partner-facing technical discussion:** provide a reusable artifact for technically grounded engagement.
- **Technical specification support:** identify which parameters deserve measurement, stabilization, or tighter tolerances.

Qunnect, Cisco, and similar ecosystem participants may serve as useful context for the broader networking discussion, but the artifact is an independent technical exploration tool.

Positioning

This sandbox should be understood as a **technical brief artifact**, not as a claim of new physics and not as a replacement for large simulation frameworks. Its value lies elsewhere:

- it is small enough to understand,
- structured enough to trust,
- technical enough to matter,
- and clear enough to support real architecture conversations.

That makes it suitable for GitHub-facing technical presentation and cross-domain, partner-facing communication.

Strategic Takeaway

As quantum networking matures, there is increasing value in artifacts that sit between textbook explanation and full-stack simulation. The Entanglement Swapping Sandbox occupies that middle layer: it turns a foundational network primitive into a compact technical tool for **scenario screening**, **tradeoff exploration**, and **specification support**.

The broader point is not simply that entanglement swapping can be simulated. The broader point is that technical concepts can be packaged into artifacts that make architecture choices easier to discuss, evaluate, and refine. The Qunnect/Cisco GothamQ demonstration (February 18, 2026) established entanglement swapping on live commercial fiber as a carrier-compatible networking primitive [2, 1].

The sandbox makes the enabling factors of that primitive — source quality, BSM alignment, loss, and classical channel latency — directly explorable, so that the next design conversation starts from a technically grounded baseline rather than a black box.

References

- [1] A. N. Craddock, T. Cowan, N. Bigagli, S. Yekasiri, D. Robinson, G. Bello Portmann, Z. Guo, M. Kilzer, J. Zhao, M. Flament, J. Shabani, R. Nejabati, and M. Namazi, “High-rate Scalable Entanglement Swapping Between Remote Entanglement Sources on Deployed New York City Fibers,” arXiv:2602.15653, 2026. <https://arxiv.org/abs/2602.15653>
- [2] Qunnect, “Qunnect and Cisco Demonstrate First Metro-Scale, High-Speed Quantum Entanglement Swapping Over Commercial Fiber,” press release, February 18, 2026. <https://www.qunnect.inc/press-releases/2026-02-18>